Few-Shot Continual Learning for 3D Brain MRI with Frozen Foundation Models

Task 1: Tumor Segmentation

Proposed Framework

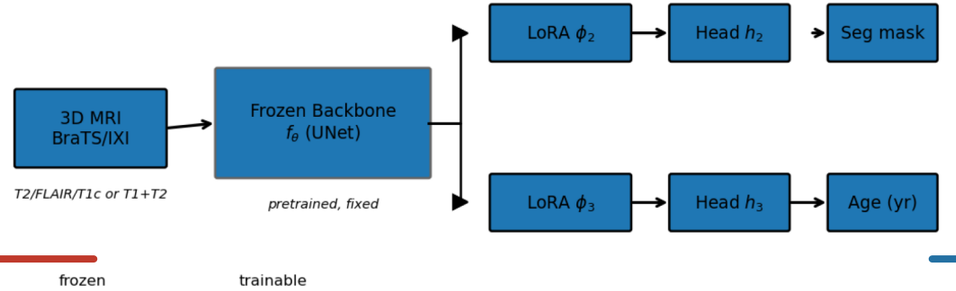
Task 2: Brain Age Estimation

Input (T2-FLAIR)

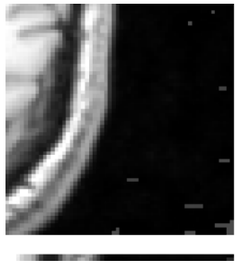


Prediction Overlay

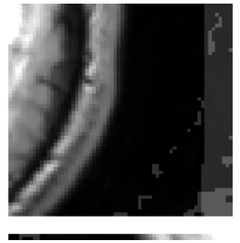




Axial View



Sagittal View



Pred: 46.0 yr | GT: 50.0 yr

BWT = 0

Zero Catastrophic Forgetting

Frozen backbone + isolated
LoRA adapters guarantee
no forgetting by design

< 0.1%

Parameter Efficient

Only ~28K-47K trainable params
per task (vs. 59M backbone)
~1.7 GB peak GPU memory

n = 32

Few-Shot Ready

Competitive performance with
as few as 32 labeled samples
per downstream task